The Hotelling Rule Holds Only on Average

Exhaustible resource prices and the real interest rate, 1960-2024

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Abstract

Over 1960-2024 the real prices of 13 exhaustible resources rose by 0.5 percent a year while the real prices of 10 renewable commodities fell by 1.7 percent a year. Hotelling's rule makes the scarcity rent, the price net of extraction cost, grow at the interest rate. If exhaustible resources share the cost declines of farm goods and the rent is a share s of their price, the gap between the two groups should be s times the real interest rate less the common cost trend. The gap of 2.2 percentage points and the average real ten-year Treasury yield of 1.9 percent imply a rent share of about 0.6. Decade by decade the relation has the wrong sign. Exhaustible prices fell 4.6 percent a year in 1980-99, when real ten-year yields averaged 4.1 percent, and rose 2.0 percent a year in 2000-24, when yields averaged 0.7 percent. Across rolling 20-year windows the exhaustible price trend has a correlation of -0.68 with the real yield, the gap over renewables has a correlation of -0.02, and annual price changes do not respond to the lagged real rate. The scarcity rent, if it exists, is visible only in the 65-year average.

1 The rule and the data

Hotelling (1931) showed that if a mine owner can extract now or later, the net price of the resource must rise at the rate of interest, since otherwise the owner would shift extraction across time. With extraction costs that fall with technology, the market price can rise more slowly or even fall, so the cleanest version of the test compares exhaustible resources with commodities that share technological progress in production and transport but carry no scarcity rent. Krautkraemer (1998) and Livernois (2009) survey a large literature that mostly fails to find the rule in individual resource prices.

We use the World Bank's annual commodity prices in nominal dollars (the Pink Sheet), 1960-2024, deflated by the US consumer price index. The exhaustible group is crude oil, Australian coal (from 1970), US natural gas, aluminum, copper, iron ore, lead, nickel, tin, zinc, gold, silver and platinum. The renewable group is maize, US hard red winter wheat, Thai 5% broken rice, soybeans, world sugar, arabica coffee, cotton (A Index), beef, cocoa and tea. Real interest rates are $\ln(1+i) - \ln(1+\pi)$, where i is the annual average three-month Treasury bill or ten-year Treasury yield and π is realized consumer price inflation over the year. Trends are least-squares slopes of log real prices on time, averaged across commodities within each group.

2 Results

Over the whole sample the rule looks right on average (Table 1). Exhaustible prices rose by 0.47 percent a year and renewable prices fell by 1.69 percent, a gap of 2.16 percentage points with a standard error of 0.35. That is close to the average real ten-year yield of 1.94 percent and larger than the real bill rate of 0.59 percent. Because the rule applies to the rent and not to the market price, the gap should equal $s(r-g)$, where s is the rent share of the price, r the real rate and $g = -1.69$ the cost trend shared with renewables, and with the ten-year yield the implied rent share is $2.16/(1.94+1.69) \approx 0.6$. The gap is not driven by oil and gold, which rose fastest. Base metals alone (aluminum, copper, iron ore, lead, nickel, tin and zinc) have a trend of -0.33 percent a year and a gap of 1.36 points over renewables.

Over shorter horizons the relation with interest rates runs the other way. Figure 1 shows that exhaustible prices fell fastest in the windows centered on the 1980s, when real rates were highest, and rose fastest in the windows centered on the 2000s, as real rates fell. Across the 46 rolling windows the correlation between the exhaustible trend and the real ten-year yield is -0.68, and the correlation between the exhaustible-renewable gap and the yield is -0.02. In annual data, with commodity fixed effects and standard errors clustered by year, the change in log real exhaustible prices on the previous year's real ten-year yield has a slope of 0.21 with a standard error of 0.83, and on the real bill rate 0.06 (0.91), against a predicted value of one. The rolling windows overlap, so the correlations describe the pattern and are not independent tests.

Frankel (2008) predicts this negative association for commodities in general. High real interest rates raise the cost of holding inventories and of leaving a resource in the ground, so owners extract faster and spot prices fall. A fall in the price level when rates rise is consistent with Hotelling, but the rule then requires prices

Period	Exhaustible	Renewable	Gap	Real T-bill	Real 10-year
1960-2024	0.47 (0.34)	−1.69 (0.10)	2.16 (0.35)	0.59	1.94
1960-1979	3.53 (0.90)	1.90 (0.72)	1.64 (1.15)	0.43	1.33
1980-1999	−4.57 (0.70)	−4.41 (0.31)	−0.16 (0.76)	2.43	4.10
2000-2024	2.00 (0.89)	1.38 (0.20)	0.62 (0.91)	−0.76	0.69

Table 1: Average trend of log real prices, percent per year, across exhaustible resources and 10 renewable commodities, with the standard error of the cross-commodity mean in parentheses, and average real interest rates in percent. A commodity enters a period if it has at least 15 annual prices in it, so Australian coal (from 1970) is excluded from 1960-1979, which averages 12 exhaustible resources; the other rows use all 13. Prices from the World Bank Pink Sheet deflated by the US CPI.

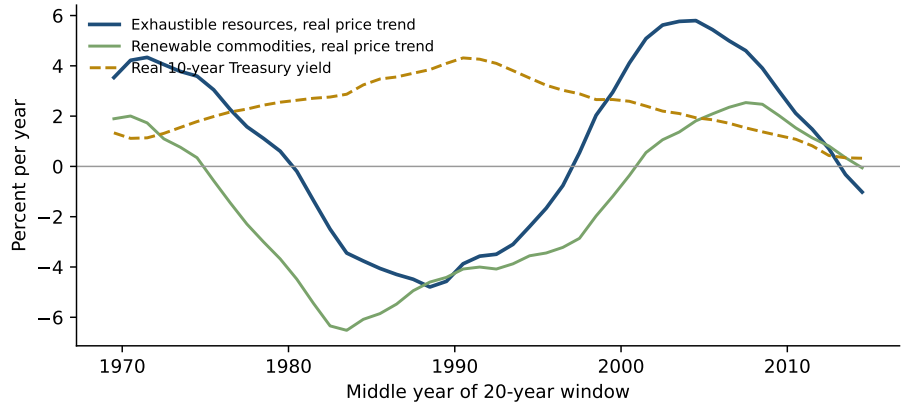


Figure 1: Average trend of log real prices in rolling 20-year windows for exhaustible and renewable commodities, and the average real ten-year Treasury yield over the same window, plotted at the middle year.

to grow faster than before, and in the 1980s they kept falling. Renewable price trends have a correlation of −0.79 with the real ten-year yield across the rolling windows, close to the exhaustible figure. The common movement therefore looks driven by the cost of carry and by demand, and a scarcity rent, if it explains the average gap, is too small to show up in the relation with interest rates over time.

3 Scope

The test uses market prices, while the rule applies to the net price after extraction costs, which are not observed. Exhaustible resources differ from farm goods in demand as well as scarcity, and the strong rise of exhaustible prices in the 2000s coincided with fast growth in China's demand for energy and metals. The average gap over 1960-2024 is therefore consistent with a Hotelling rent, but it does not identify one. Harvey, Kellard, Madsen and Wohar (2010) document the long-run decline of relative primary commodity prices over four centuries.

References

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